There Is No Unmatrix: What Simulation Changes and What It Does Not

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Abstract

If future civilizations produce vastly more minds, why do we seem to live early? The number of future minds alone cannot give the answer. A probability calculation also needs a rule for which observers count and how they are weighted. A worked example shows that changing this rule changes the predicted odds. A later simulation could also contain a mind with memories of an earlier age. On Matheism, such a mind is still part of mathematical reality. The distinction matters for simulation probabilities without placing simulated experience outside reality.

Keywords: algorithmic probability, temporal location, observer-moment, doomsday argument, consciousness, artificial intelligence, Kolmogorov complexity

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1. The Question

If you could be born at any point in the history of sentience, when would you expect to find yourself?

Suppose the future contains many more minds than the present. Does that make our apparently early position unlikely? Only after we decide which minds or moments to count, and how much weight each receives. A simulated mind could also remember an earlier era. Our apparent date and the date of the machinery supporting us are different questions.

1.1 Development does not determine our temporal rank

Simple laws can remain unchanged while the structures they generate acquire the organization required for observers. Suppose the starting state lacks that organization and a later state contains observers who require it. The history must then develop the missing organization. This is a necessary consequence of those premises, not merely a possible outcome. The growing quantity is the observer relevant organization of the generated states; the law and the shortest description of the history can remain simple.

Reaching the threshold establishes that the required development occurred. Determining whether we should be early, late, simulated or near a peak additionally

requires weights over observer situations. A count of possible future observers alone does not settle that comparison.

2. Three Readings

2.1 Simulation

A computer that develops under a specified law and boundary is conditionally describable through that history. Its conditional description may be short, but chemistry, life, technology and particular simulations are not guaranteed by the laws alone. The program, resources and preparation needed to implement a chosen simulation must be identified. Do not charge them twice when the complete history already generates them; do not omit them when it does not.

A sufficiently large population of simulated observers could dominate a specified observer measure. The conclusion is conditional on how simulations are implemented, what experiences they support and how their instances are weighted. A bare count of possible simulations does not establish that we are probably simulated.

If our world is simulated, a programmer could be a creator within a larger reality. Under mathematical monism that programmer is structure too. Another matrix. No unmatrix. The interpretation changes the causal setting of our world without supplying evidence that this setting is actual.

2.2 Sentience Near Its Typical Peak

One candidate is that technological expansion is uncommon or limited. Its relevance to our location depends on the frequencies, durations and observer weights of those histories. A history with many minds early and few later can make early observations typical under one sampling rule, but the rule and the population model must be stated.

Rahvar and Rouhani (2026) derive a lifespan bound of roughly 5,000 years in their most optimistic scenario for the emergence of intelligent life. They explicitly describe it as a conditional upper bound, not a prediction of actual lifespans. Nondetection and apparent temporal rank are different evidence. Combining them requires a joint likelihood, including detectability and observer selection; they cannot be counted as two independent confirmations merely because both concern civilization duration.

2.3 The Transition

Another candidate is a change in the forms or prevalence of experience. Future computation might support many minds, few minds or none. Shutdown and integration could alter subject counts, but counting systems is not yet counting observer moments. Those scenarios require criteria for sentience, subject identity and duration.

One integrated subject could contain an enormous organization and undergo many distinguishable experiences. The word one therefore implies neither negligible Kolmogorov complexity nor one observer moment. A merger hypothesis needs its own account of which experiences continue and how they are weighted.

Planet and asteroid counts do not bound the number of computational observers: several systems could share one body, and habitats need not occupy natural surfaces. Communication delays constrain interaction but do not determine subject identity. A useful population model specifies available energy, computation, duration and a consciousness criterion before drawing a temporal ranking.

Recognizing common structure can motivate concern for other minds. It does not by itself determine a utility function, require merger or prevent conflicting interests. Ethical conclusions about a unified system need premises about its experiences and values.

3. Why There Is No Unmatrix

Mathematical monism identifies a mind with its complete structure. On this view, biological development, artificial construction and simulation can differ in implementation without requiring distinct kinds of substance.

If an implemented simulation is itself embedded in another simulation, the structural account applies at each level. There is no need within this ontology to add a nonmathematical final substrate. That is the sense in which there is no unmatrix; it is not a probability calculation establishing that our world has a programmer.

A simulated world could reflect a creator's choices, including choices relevant to suffering. This is a conditional interpretation, not evidence for a creator or its motives. The monist claim concerns what the creator would be, not whether one is present.

4. Implications for AI

Each reading bears on AI development.

Under an extinction model, evaluate the physical mechanisms and their likelihoods. Early temporal location does not establish extinction as the default outcome without a specified observer measure.

Under the wild card: AI consciousness arises but remains local. No expansion, no merge. Many civilizations, each with their own AI minds, none colonizing. The future looks like the present at scale.

Under a merger model, ask whether integration changes the number of subjects, the content of experience or merely communication between existing subjects. No general inference from connectivity to one awareness is established here.

These scenarios do not jointly predict that a transition is imminent. A timing prediction needs a distribution over transition dates and a likelihood for our observations.

5. Relation to Existing Work

Bostrom (2002) distinguishes observer sampling assumptions that can produce different conclusions. A prior over programs does not eliminate that choice. Leuenberger (2014, revised 2017) already combines algorithmic probability with observer localization in applications including simulation. The present paper's contribution is the explicit separation of history weight, observer location and apparent date, illustrated by the finite comparison below. Chalmers (2022) addresses the reality of simulated worlds, a question distinct from their probability.

Philosophical proposals about shared identity, including Kolak and Zuboff, can motivate alternative subject criteria. They do not supply numerical observer weights simply by identifying one subject. The probability analysis must state how it individuates experiences across time, copies and implementations.

Worked comparison: the sampling rule changes the answer

Take two finite histories with equal prior weight. Hs contains ten early and ten late observer moments. Hl contains ten early and one thousand late moments. Assume for this example that all moments within a history receive equal weight and that early status is known accurately.

History	Early moments	Late moments	Total	Chance of early status within that history
Hs	10	10	20	1/2
Hl	10	1000	1010	1/101

Under sampling within each history, Bayes's rule gives P(Hl | early) = (1/101)/(1/2 + 1/101) = 2/103, about 1.94 percent. Under a second rule that first weights histories by their total observer count, conditioning on early status gives weights proportional to 10 and 10, hence P(Hl | early) = 1/2. The prior over descriptions did not change; the observer weighting did.

More generally, let w(H) be a history weight and q(j | H) a normalized location weight over its moments. Then P(D | H) = sum_j q(j | H) P(D | H,j). A count weighted alternative first multiplies w(H) by a specified total observer weight. Neither choice follows merely from taking w(H) proportional to 2^(−L(H)). The finite example is a demonstration of this dependence, not a new solution to anthropic probability.

For simulations, D should describe the observer's accessible evidence. A late simulation of an early era can generate early looking records. External date and represented date therefore cannot be identified without argument.

The positive simulation argument can now be stated. Within a specified history, suppose m implemented simulated observer moments and b other observer moments receive equal individual weight and assign the same likelihood to our evidence D. Then P(simulated | D,H) = m/(m+b). One hundred such simulated moments for every other moment gives 100/101. This is the numerical force of the population argument under its assumptions. The physical and philosophical tasks are to justify the counts, implementation criterion, evidence likelihoods and chosen observer weights; comparing histories then includes their prior weights too.

6. Conclusion

A simulated experience can be real even when its represented surroundings are fictional. Mathematical monism gives both simulated and other minds the same proposed kind of reality.

To estimate whether we are simulated, counts need an explicit rule and comparison with our records. The worked example shows how that choice changes the answer. A large future population matters through the particular experiences being counted and the weights assigned to them.

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